Lecture 28 — Eigenvectors as persistent directions

Mon Nov 16

Last updated: October 29, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §6.1 (PDF) — eigenvalues
Video: 18.06 Lec 21 — eigenvalues and eigenvectors · MIT 18.06 playlist (Strang)

Reading: Beginning of Strang §6.1 — $Ax = \lambda x$, eigenvectors, and dynamics. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Story of the eigen week.

Day	Question
Lab 4 (done)	Why do different campus starts settle to the **same** long-run mix?
Today (L28)	Name the idea: **eigenvectors** = directions that only get rescaled
L29 Wed	How to **find** eigenpairs by hand (2×2 + complex)
L30 + Lab 5 Fri	Explain Lab 4 with software: $\lambda = 1$ wins, other modes fade

Learning objectives

By the end of class, you should be able to:

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair.
3. Find λ when an eigenvector is given.
4. Connect $A^t v = \lambda^t v$ to repeated dynamics (powers from Lecture 23).
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.

Warm-up

Lab 4's campus matrix (same every week):

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

You saw $x_0 = (1000, 0)$ and $x_0 = (0, 1000)$ both head toward about $(400, 600)$.

(a) In one sentence: what was surprising?

(b) Preview: if some nonzero v satisfied $Av = v$ (eigenvalue 1), why would iterating from that v stay put (up to scaling of total population)?

Fill in

Eigenpair. A nonzero vector v is an **eigenvector** of square A with **eigenvalue** λ if

$$Av = _____.$$

Geometrically: A sends v to a vector on the _____ line through v (scaled by λ; if $\lambda < 0$, also flipped).

Verify. Given a candidate pair (λ, v), compute Av and λv; they must _____.

Find λ from v. If $v \neq 0$ is known to be an eigenvector and some component $v_i \neq 0$, then $$